American Computer Science League

2024 Finals ● Short Problems Solutions ● Intermediate/Classroom Division

1. Boolean Algebra

$A(\overline{B}C) + \overline{ABC} + (\overline{A} + \overline{B})C = A\overline{B}C + \overline{A} + \overline{B} + \overline{C} + \overline{A}C + \overline{B}C$

$= \overline{A}(1 + C) + \overline{B}(AC + \overline{1} + C) + \overline{C}$

$= \overline{A} + \overline{B} + \overline{C}$

$\overline{A} + \overline{B} + \overline{C} = 0 \rightarrow \overline{A} = 0 \wedge \overline{B} = 0 \wedge \overline{C} = 0$

So $A = 1, B = 1, C = 1$.

Therefore only (1,1,1) makes it False.

B. 1

2. Boolean Algebra

$\overline{\overline{B}A} + (B \oplus C) = \overline{\overline{B}} + \overline{A} + \overline{B}C + B\overline{C}$

$= B(1 + \overline{C}) + \overline{A} + \overline{B}C$

$= \overline{A} + B + \overline{B}C$

D. $\overline{A} + B + \overline{B}C$

3. Bit-String Flicking

01100 OR (LSHIFT-1 (RCIRC-2 01011))
AND (LCIRC-2 (RSHIFT-1 01010))
= 01100 OR (LSHIFT-1 11010) AND (LCIRC-2 00101)
= 01100 OR (10100 AND 10100)
= 01100 OR 10100
= 11100

C. 11100

4. Bit-String Flicking

(RCIRC-2 01101) AND (LSHIFT-1 X) =
(LSHIFT-2 (NOT 10111 OR X))

Let X = abcde.

LHS = (RCIRC-2 01101) AND (LSHIFT-1 X)
= (RCIRC-2 01101) AND (LSHIFT-1 abcde)
= 01011 AND bcde0
= 0c0e0

RHS = (LSHIFT-2 (NOT 10111 OR X))
= (LSHIFT-2 (01000 OR abcde))
= (LSHIFT-2 a1cde)
= cde00

LHS = RHS → 0c0e0 = cde00
a = *, b = *, c = 0, d = 0, e = 0

Therefore there are 4 solutions: **000

C. 4

5. Recursive Functions

$f(18) = f(18-3) + 1 = f(15) + 1 = 33 + 1 = 34$
$f(15) = f(15-3) + 1 = f(12) + 1 = 32 + 1 = 33$
$f(12) = 2 * f(12+1) - 2 = 2 * f(13) - 2 = 2 * 17 - 2 = 32$
$f(13) = f(13-3) + 1 = f(10) + 1 = 16 + 1 = 17$
$f(10) = 2 * 10 - 4 = 16$

E. None of the above

6. Recursive Functions

$f(40, 28) = f([40/3], [28/2]) + 1 = f(13, 14) + 1 = 13$
$f(13, 14) = f(14+4,13-2) = f(18,11) = 12$
$f(18, 11) = f([18/3],[11/2])+1 = f(6, 5)+1 = 12$
$f(6, 5) = 6 + 5 = 11$

D. 13

7. Digital Electronics

The Boolean expression for the circuit is: $(AB) \oplus (B + C)$

$(AB) \oplus (B + C) = \overline{AB}(B + C) + AB(\overline{B + C})$
$= (\overline{A} + \overline{B})(B + C) + A B \overline{B}\,\overline{C}$
$= \overline{A}B + \overline{A}C + \overline{B}B + \overline{B}C + 0$
$= \overline{A}B + \overline{A}C + \overline{B}\,C$

$If\ \overline{A} = 1,\ then\ A = 0\ \wedge\ B + C + \overline{B}C = 1$
$If\ B = 1,\ then\ 1 + C + 0 = 1 \rightarrow$ C=* (0,1,*)
$If\ B = 0,\ then\ 0 + C + C = 1 \rightarrow C = 1$ (0, 0, 1)
$If\ \overline{A} = 0,\ then\ A = 1\ \wedge\ 0 + 0 + \overline{B}C = 1$
$If\ C = 1,\ then\ \overline{B} = 1 \rightarrow B = 0$ (1, 0, 1)
$If\ C = 0,\ then\ 0 = 1\ which\ is\ impossible.$
Therefore 4 ordered pairs make it TRUE.

B. 4

8. Digital Electronics

The Boolean expression for this circuit is: $\overline{(A + \overline{AB}) + (\overline{B + C}\,\overline{C})}$

$\overline{(A + \overline{AB}) + (\overline{B + C}\,\overline{C})} = \overline{(A + \overline{A} + \overline{B}) + (\overline{B}\,\overline{C}\,\overline{C})}$
$= \overline{1 + \overline{B} + \overline{B}\,\overline{C}}$
$= \overline{1}$
$= 0$

A. 0

9. Prefix-Infix-Postfix

\# − * # 2 ^ 3 2 $ * 4 2 3 − ^ 6 2 * 3 4 0
= # − * # 2 (^ 3 2) $ (* 4 2) 3 − (^ 6 2) (* 3 4) 0
= # − * (# 2 9) ($ 8 3) (− 36 12) 0
= # − (* 2 8) 24 0
= # (− 16 24) 0
= # -8 0
= -8

C. -8

10. Prefix-Infix-Postfix

$$\frac{5(2A+4)}{2B-3} - 7$$

$= (5\ *\ (2\ *\ A + 4))/(2\ *\ B - 3) - 7$
$= (5\ *\ (2\ A\ *\ 4\ +))/(2\ B\ *\ 3\ -) - 7$
$= (5\ 2\ A\ *\ 4\ +\ *) / (2\ B\ *\ 3\ -) - 7$
$= (5\ 2\ A\ *\ 4\ +\ *\ 2\ B\ *\ 3\ -\ /) - 7$
$= 5\ 2\ A\ *\ 4\ +\ *\ 2\ B\ *\ 3\ -\ /\ 7\ -$

A. 52A*4+*2B*3−/7−

11. Computer Number Systems

$77777_8 + 7777_8 - 777_{16} = (77777_8 + 7777_8) - 777_{16}$
$= 107776_8 - 777_{16}$
$= 8FFE_{16} - 777_{16}$
$= 8887_{16}$

C. 8887

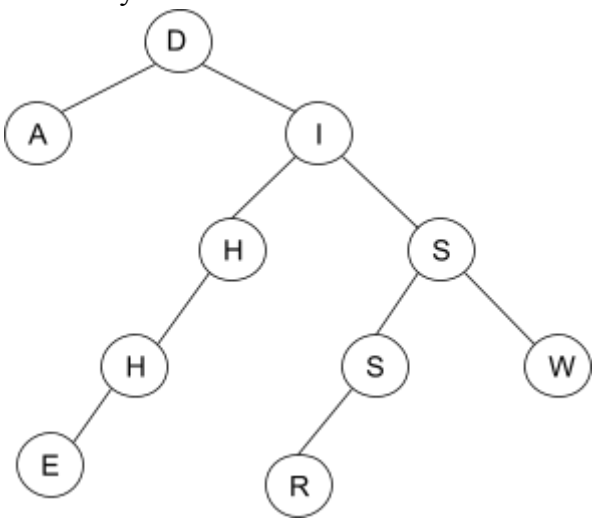
12. Computer Number Systems

$3_8 \wedge 1 = 3$
$3_8 \wedge 2 = 9_{10} = 11_8$
$3_8 \wedge 3 = 27_{10} = 33_8$
$3_8 \wedge 4 = 81_{10} = 121_8$
Either continue to 7th power or recognize the pattern.
If the power is odd, the last octal digit is 3.
If the power is even, the last octal digit is 1.
Since 7 is odd, the last octal digit is 3.

C. 3

13. Data Structures

The binary search tree for **DISHWASHER** is:



The preorder trasversal is: DAIHHESSRW

D. DAIHHESSRW

14. Data Structures

This algorithm doesn't check to see if the symbols match (i.e. {}, [], ()), only if there are the same number of open and closed symbols in the right order. Therefore, C creates an empty stack, but D doesn't.

D. {[(x-2y[(3x+4]](2x]}

15. Graph Theory

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}^4 = \begin{bmatrix} 0 & 2 & 2 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

There are 7 paths from A, 4 from B, 2 from C, and 3 from D.
A has the most paths of length 4.

A. A

16. Graph Theory

There are 6 cycles in the graph. They are ABEDA, ABEDCA, AEDA, AEDCA, BEB, and DED.

D. 6

17. What Does This Program Do?

The value of s1 is the sum of the sequence 1+4+9+16+25+36+... which is the sequence 1, 5, 14, 30, 55, 91, ... while the value of s2 is every triangular number 1+2+3+4+5+6+... which is the sequence 1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78, 91. They are the same for the 10th and the 13th triangular numbers. The value of 1 isn't compared because the loop starts with 2.

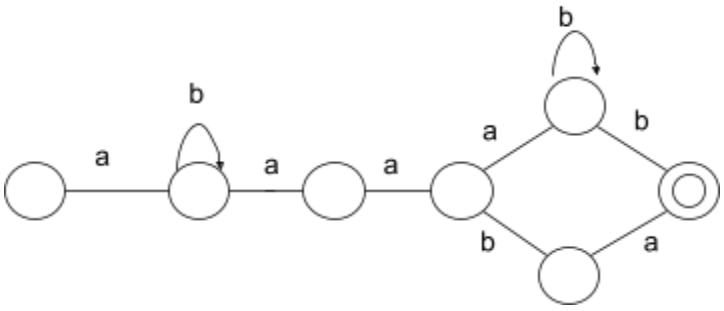
C. 10, 13

18. LISP

(REVERSE (CADADR '(1 (2 (3 2 (1))) (2 ((2) 1)))))
= (REVERSE (CADAR '((2 (3 2 (1))) (2 ((2) 1)))))
= (REVERSE (CADR '(2(3 2 (1)))))
= (REVERSE (CAR '((3 2 (1)))))
= (REVERSE '(3 2 (1)))
= ((1) 2 3)

A. ((1) 2 3)

19. FSAs



The regular expression for this FSA is: ab*aa(ab*bUba)

- a b b a a b fails at third b
- a a a b a satisfies it
- a b a a b a b fails at third b
- a b b b a a a needs a b at the end
- a b b a a b a a has one too many as at end

B. exactly 1 of them

20. Assembly Language

This program converts the binary number to base 10 by getting each digit from the right to the left using integer division and remainders. Each digit, a 0 or 1, is multiplied by increasing powers of 2 which is 2+4+8+32+128=174.

C. 174